

# 1. Tic - Tac - Toe

O → human

X → AI agent

$$S_0$$

|   |   |   |
|---|---|---|
| 0 |   | X |
| X |   |   |
| X | 0 | 0 |

$$S_1$$

|   |   |   |
|---|---|---|
| 0 | X | X |
| X |   |   |
| X | 0 | 0 |

$$S_2$$

|   |   |   |
|---|---|---|
| 0 |   | X |
| X | X |   |
| X | 0 | 0 |

$$S_3$$

|   |   |   |
|---|---|---|
| 0 |   | X |
| X |   | X |
| X | 0 | 0 |

$S_4$

$S_5$

$S_6$

$S_7$

$S_8$

$S_9$

|   |   |   |
|---|---|---|
| 0 | X | X |
| X | 0 |   |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | X | X |
| X |   | 0 |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | 0 | X |
| X | X |   |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 |   | X |
| X | X | 0 |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | 0 | X |
| X |   | X |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 |   | X |
| X | 0 | X |
| X | 0 | 0 |

$S_{10}$

$S_{11}$

$S_{12}$

$S_{13}$

$S_{14}$

$S_{15}$

|   |   |   |
|---|---|---|
| 0 | X | X |
| X | 0 | X |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | X | X |
| X | X | 0 |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | 0 | X |
| X | X | X |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | X | X |
| X | X | 0 |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | 0 | X |
| X | X | X |
| X | 0 | 0 |

|   |   |   |
|---|---|---|
| 0 | X | X |
| X | 0 | X |
| X | 0 | 0 |

win(human)  
(1)

win(AI)  
(1)

win(AI)  
(1)

win(AI)  
(1)

win(AI)  
(1)

win(human)  
(1)

## Pseudocode

Step 1:- Start the game

Step 2:- Create 3x3 grid

Step 3:- Human place an O and AI places an X

Step 4:- While game is not completed

Step 5:- Human selects an empty cell and AI selects an empty cell.

Step 6:- Check 3 symbols same in same row, same column and diagonal.

Step 7:- The one who has same symbol in same row, same column and same diagonal wins the game

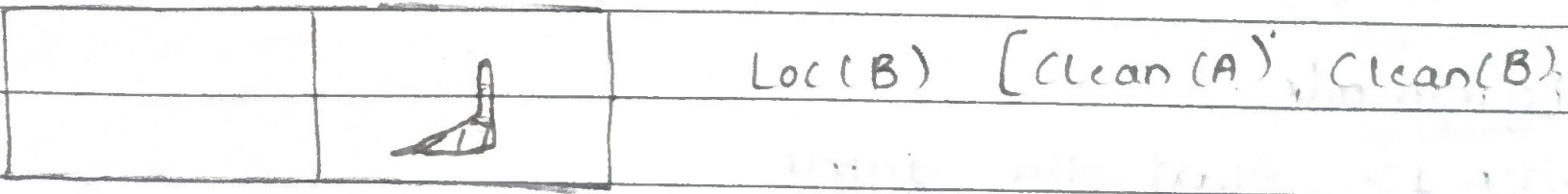
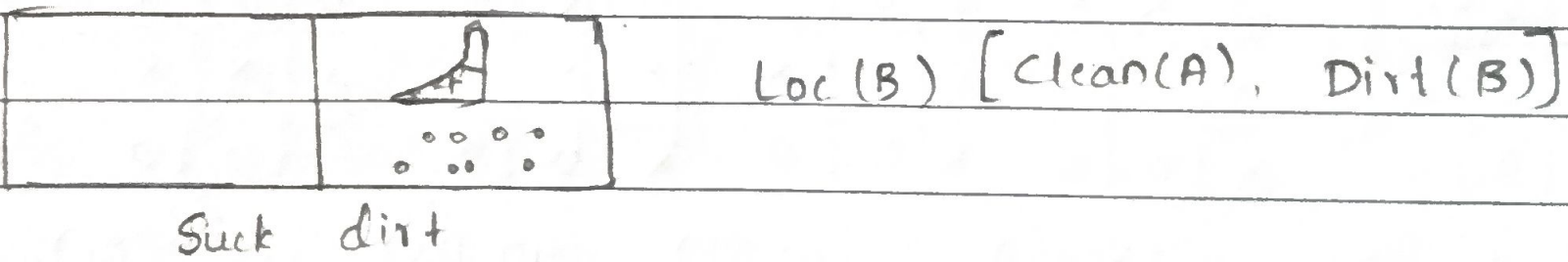
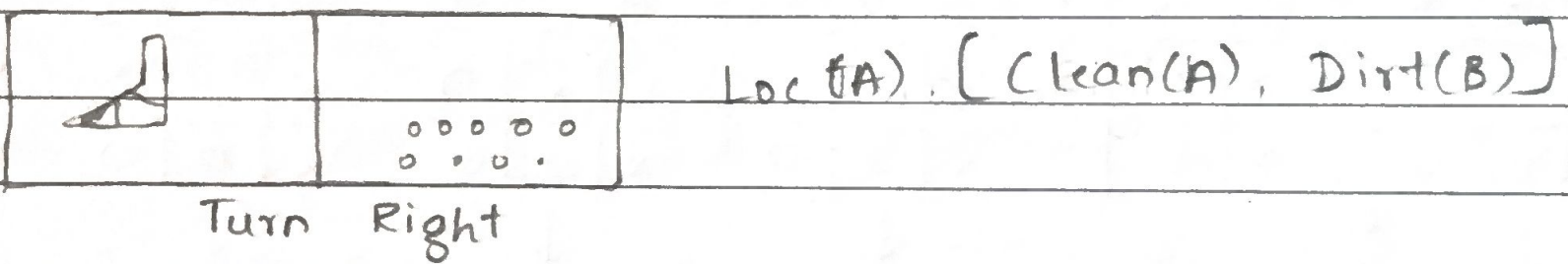
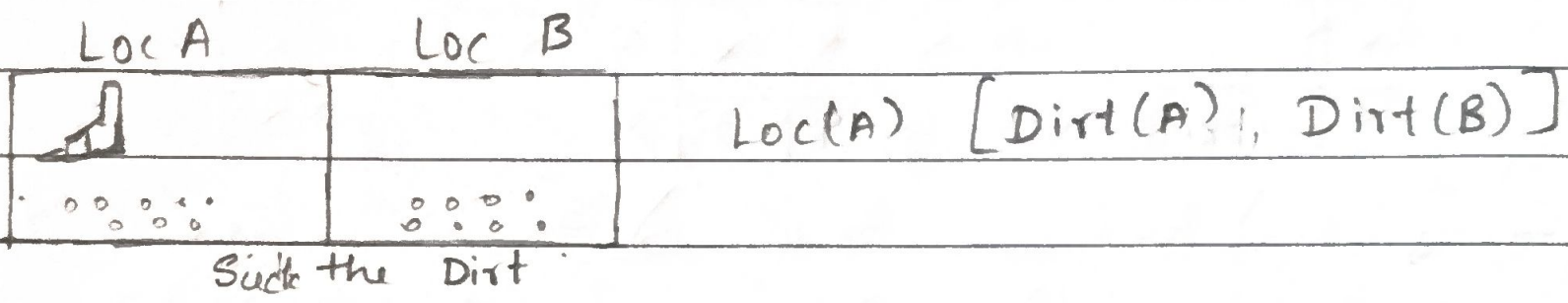
Step 8:- If no one wins, the game is draw

Step 9:- Stop.

2. Vacuum Cleaner Agent

Function REFLEX\_VACUUM\_AGENT ([location, status]) returns an action

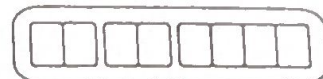
If status = Dirty then return pick the Dirt  
else if location = A then return Right  
else if location = B then return Left.



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# Tic-tac-toe

## Output



Welcome to Tic-Tac-Toe!

You are 'X' and the AI is 'O'.

(Choose) a position (0-8): 0

|   |  |  |
|---|--|--|
| X |  |  |
|   |  |  |
|   |  |  |

AI's turn

|   |   |  |
|---|---|--|
| X |   |  |
|   | O |  |
|   |   |  |

Choose a position (0-8): 2

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

|   |   |   |
|---|---|---|
| X |   | X |
|   | O |   |
|   |   |   |

AI's turn

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

Choose a position (0-8): 7

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

|   |   |   |
|---|---|---|
| X |   | X |
|   | O |   |
|   |   | X |

AI's turn

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

Choose a position (0-8): 5

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

|   |   |   |
|---|---|---|
| X |   | X |
|   | O |   |
|   |   | X |

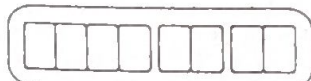
AI's turn

|   |   |   |
|---|---|---|
| X |   |   |
|   | O |   |
|   |   | X |

Choose a position (0-8): 6

|   |   |   |
|---|---|---|
| X |   | X |
|   | O |   |
|   |   | X |

It's a tie.



## Vacuum Cleaner Agent (Output)

### --- Vacuum Cleaner Agent Simulation ---

Enter current vacuum location (A or B): A

Enter status of Location A (Clean or Dirty): Dirty

Enter status of Location B (Clean or Dirty): Dirty

Initial State: Vacuum is at A. Room A is Dirty,  
Room B is Dirty

[Step 1] Location: A

Status: Dirty → Action Chosen: Suck

--> Success: Location A has been cleaned

Current Environment Status: Room A: Clean

Room B: Dirty

[Step 2] Location: A

Status: Clean → Action Chosen: Right

--> Success: Move Right to location B

Current Environment Status: Room A: Clean

Room B: Dirty

[Step 3] Location: B

Status: Dirty → Action Chosen: Suck

--> Success: Location B has been cleaned

Current Environment Status: Room A: Clean

Room B: Dirty.

*Handwritten signature/initials*